

Integration of Small Pumped Storage Hydropower Units in Water Distribution System Operation

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Abstract—Pumped-storage hydropower (PSH) is a proven energy storage technology that can provide large capacity support to the bulk power system. PSH is also a promising technology to increase energy storage capabilities of water distribution systems (WDSs), because these systems have most components necessary for supporting PSH, including a water supply, conduit, pumps, and water storage capacity. PSH must be sized to the water infrastructure that the energy storage will be integrated with, but generally these PSH applications are significantly smaller than traditional PSH units. This paper develops an optimization framework for integration of small PSH units in WDS operation to minimize operating costs associated with power and water consumption in WDS. The proposed model takes into account the hydraulic constraints of WDS and provides optimal operating strategies from an economic perspective based on available units in WDS (e.g., the hydropower generator, variable speed water pumps, and utilization of water storage capacity). The proposed model is applied to a 16-node test WDS using two cases that differ in terms of PSH design to demonstrate the merits of the proposed model. The results provide insight into improved operations for the test system. They also provide the basis for improving operations of more complex WDS systems, including enabling evaluation of PSH design and integration.

Index Terms—Water distribution systems, small pumped-storage hydropower (PSH), energy management, optimization.

I. INTRODUCTION

Energy storage represents the most promising solution to compensate for the variability and uncertainty of renewable generation in power systems. Pumped-storage hydropower (PSH) is a proven energy storage technology that can provide a wide range of storage requirements depending on its design, from short-time scale frequency regulation to multi-hour load following [1], [2]. Historically most PSH plants have had large capacities (on the order of 100's of MW) to provide centralized storage for the bulk power system. There has been a growing discussion, though, about alternative PSH designs and operating paradigms to meet specific power system needs and enable greater development of PSH technologies. Some studies demonstrated that local, small applications of PSH units can support integration of renewable distributed energy resources (DERs), and may be more suitable compared to the large-scale types implemented in bulk energy systems [3]. For this reason, there has been efforts to design small PSH units with capacities in kW to MW range for specific applications and to utilize existing infrastructure [4]–[6].

Integration of small PSH units with DERs can result in efficient hybrid energy systems. These systems can serve energy demands in multiple sectors and provide positive social impacts as the community can have access to both electricity and water resources [7]–[9]. One application of such small PSH units integrated in hybrid energy systems is in rural and remote areas. In such areas, the community may have problems with electricity reliability [10], [11]. Also, DERs that are usually renewable generation resources like photovoltaic (PV) and wind, along with small PSH units can be integrated in hybrid energy systems to supply a sustainable and continuous power to the consumers [12]–[14]. The community living in rural regions can also benefit from such integrated units for farming purposes [15]–[17]. In [18], PSH unit is integrated into a renewable-based on-grid microgrid irrigation system to serve water and electricity demands under a market concept in different levels i.e. industrial, lighting. It should be noted that implementation of small PSH units is not limited to remote and rural areas. Recently, some studies assessed the impact of implementing small PSH units in building levels. Authors in [19], [20] studied implementation of small PSH units for tall buildings and analyzed the feasibility of presented models from economic, technical and social perspectives.

One available and efficient solution to develop PSH in water networks is to convert the current water pumps into storage systems. Pumping units can be redesigned to result in PSH units with the capability of pumping water and generating power. This issue is studied in large water supply networks in [21]. However, in water distribution systems (WDSs) with small water flow rates, large-scale PSH systems may not be technically and economically feasible to be integrated. Therefore, small PSH units can be appropriate candidates to be developed and utilized in WDSs. In order to serve water demands, WDSs are usually equipped with specific facilities like water tanks and water pumping motors [22]–[24] that can be redesigned to create small PSH units. This requires that the operating characteristics of the pumps be proportional with the structure of WDS (e.g., the pipes, water availability, and storage capacity).

This paper proposes a novel optimization model to integrate small PSH units into WDS systems operation that benefit from pumping motors and water tanks to serve water demands. The structured of the proposed model, including the system components and input data are shown in Fig. 1. In the proposed

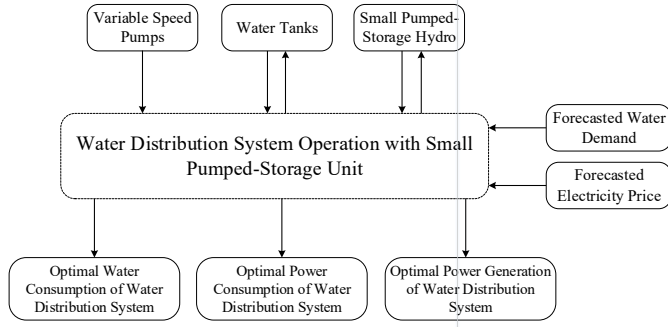


Fig. 1: Structure of the proposed WDS operation model

model, WDS operator seeks to utilize PSH to reduce the operation cost of WDS and gain economic benefit. In addition, the proposed model optimizes the operation of water pumps and tanks with the small PSH units to ensure that the hydraulic constraints of WDS are satisfied.

The rest of this paper is organized as follows: The structure of small PSH units, and the proposed optimization model for integrating small PSH into WDS's operation are presented in Section II. Numerical studies are presented in Section III and the conclusions are drawn in Section IV.

II. OPTIMAL OPERATION OF WATER DISTRIBUTION SYSTEMS WITH PUMPED STORAGE HYDROPOWER

A. Small Pumped-Storage Hydropower Units

Different types of storage technologies, with different operational paradigms and storage mechanisms, are used to improve performance of different energy systems. For example, in a WDS, water with gravitational potential or pressure differential is a natural mechanism for storing energy; whereas in a gas network, pressurized gas is a more natural energy storage mechanism. Therefore, specific storage designs and corresponding operational paradigms are needed for a given system. Water is utilized in many systems and PSH is a relatively effective method of storing energy. PSH accounts for most of the existing installed grid scale capacity. However, in recent decades there has been no new grid-scale PSH development in the United States, due to various factors including challenges acquiring land and water rights, long permitting processes, and the large-scale nature of the projects.

In recent years, there has been increased interest in smaller PSH technologies that are right-sized for integration in specific water and power systems and that overcome many of the aforementioned challenges. Small PSH systems can be designed to integrate directly with water systems such as municipal water supplies, irrigation infrastructure, or industrial processes. Small PSH systems can optimize the operation energy systems and enable them to operate more economically and reliably. These storage systems can also provide flexibility to the power grid to improve their performance. Due to the ubiquity of water, there are a large range of systems for which small PSH technology may be appropriate, including the systems in rural and urban areas, buildings, and many other forms of

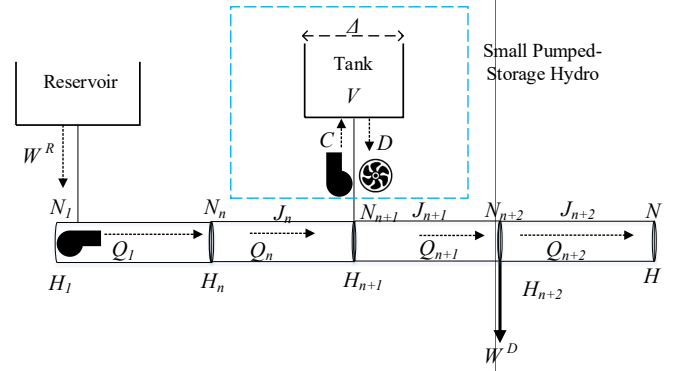


Fig. 2: Integration of small pumped-storage hydropower in water distribution system

WDSs. Small PSH units should be designed to utilize existing infrastructure of WDS and manage the energy consumption in accordance with energy tariff.

The design of a small PSH depends on the application, but in general, it is composed of different components such as a pump, tank and hydropower generator. The operating paradigm for an integrated PSH system is to meet existing water use commitments, pump water in low-price periods, store the pumped water, and generate electricity in high-price periods for behind the meter use or sale to the power grid. Integration of small PSHs in WDSs is schematically shown in Fig. 2. In a PSH integrated with WDS, water is pumped from nodes with low pressure head to water tanks with sufficient pressure head to create generation potential. This stored water is released at the desired time to meet water demands and generate electricity by the hydropower generator.

B. The Proposed WDS Operation Model

Here we present the proposed model for integrating small PSH in WDS operation. In the proposed model, formulated in (1)-(17) as a mixed-integer non-linear programming (MINLP) problem, WDS operator seeks to minimize daily operation cost of the system in (1), while satisfying hydraulic limitations and operating constraints of the water network and PSH units in (2)-(17):

$$\min \sum_{t=1}^{NT} \left(\mathbf{W}_t^{RS} \Theta \lambda^T + \mathbf{1}_U^T \mathbf{P}_t^{Pump} \pi_t^D - \mathbf{1}_M^T \mathbf{P}_t^{PSH} \pi_t^D \right), \quad (1)$$

$$\text{s.t. } \mathbf{W}_t^{RS} = \mathbf{A} \mathbf{Q}_t + \mathbf{\Lambda}^T (\mathbf{C}_t - \mathbf{D}_t) + \mathbf{W}_t^d, \quad (2)$$

$$\mathbf{B} \mathbf{A}^T \mathbf{H}_t = \mathbf{\Omega}_t^2 \odot \left(a \mathbf{1}^U - (b (\mathbf{1}^U \odot \mathbf{\Omega}_t) \odot \mathbf{B} \mathbf{Q}_t)^c \right), \quad (3)$$

$$\eta^P \mathbf{P}_t^{Pump} = \rho g (\mathbf{B} \mathbf{A}^T \mathbf{H}_t) \odot \mathbf{Q}_t, \quad (4)$$

$$0 \leq \mathbf{P}_t^{Pump} \leq \bar{\mathbf{P}}^{Pump}, \quad (5)$$

$$\mathbf{B} \mathbf{Q}_t \geq 0, \quad (6)$$

$$\mathbf{V}_t = \mathbf{V}_{t-1} + \mathbf{C}_t - \mathbf{D}_t, \quad \forall t > 1, \quad (7)$$

$$\mathbf{V}_{t=1} = \mathbf{V}_{ini}, \quad (8)$$

$$\mathbf{V}_t = \Delta \mathbf{A} \mathbf{H}_t, \quad (9)$$

$$\underline{\mathbf{V}} \leq \mathbf{V}_t \leq \overline{\mathbf{V}}, \quad (10)$$

$$\underline{\mathbf{C}} \leq \mathbf{C}_t \leq \overline{\mathbf{C}}, \quad (11)$$

$$\underline{\mathbf{D}} \leq \mathbf{D}_t \leq \overline{\mathbf{D}}, \quad (12)$$

$$\mathbf{P}_t^{PSH} = \eta^H \rho g \mathbf{H}_t^{av} \odot \mathbf{Q}_t^{PSH}, \quad (13)$$

$$-\mathbf{A}^T \mathbf{H}_t = \mathbf{G} \Pi(\mathbf{Q}_t) |\mathbf{Q}_t|^{1.852}, \quad (14)$$

$$\underline{\mathbf{H}} \leq \mathbf{H}_t \leq \overline{\mathbf{H}}, \quad (15)$$

$$\Theta \mathbf{H}_t = \mathbf{Z}, \quad (16)$$

$$\underline{\mathbf{Q}} \leq \mathbf{Q}_t \leq \overline{\mathbf{Q}}. \quad (17)$$

The objective function, (1), minimizes the total operation cost of WDS including water procurement cost and power consumption cost of water pumps, and also includes the benefits from the revenue of generating power by PSH, where $\mathbf{W}_t^{\text{RS}}$ is an $(N \times 1)$ dimensional vector of water purchased from reservoirs and supplied to the water nodes, λ is an $(R \times 1)$ dimensional vector of water price, $\mathbf{P}_t^{\text{Pump}}$ is power consumption of pump, $\mathbf{P}_t^{\text{PSH}}$ is power generated by the hydropower generator of PSH unit and π_t^D is the electricity price. The hydraulic water balance constraint in WDS is expressed in (2), where $\mathbf{Q}_t$ is a $(J \times 1)$ dimensional vector of volumetric flow, $\mathbf{A}$ is an $(N \times J)$ dimensional node-arc incidence matrix, $\mathbf{C}_t$ is input water to the tank, $\mathbf{D}_t$ is the water released from the tank, $\mathbf{A}^T$ is a $(T \times N)$ dimensional tank-node matrix and $\mathbf{W}_t^d$ is water demand. The pressure head and power consumption of pumps are calculated in (3)-(4), where $\mathbf{B}$ is a $(U \times J)$ dimensional pump-arc incidence matrix, $\mathbf{H}_t$ is a $(N \times 1)$ dimensional vector of pressure head of the nodes, Ω_t is a $(U \times 1)$ dimensional vector of pump speed, a and b are specific pump parameters, η^P is the efficiency of pumps, ρ is water density and g is acceleration due to gravity. Power consumption of the pump is constrained by (5), where $\overline{\mathbf{P}}^{\text{Pump}}$ is the rated power of pump. Constraint (6) ensures that pumping is done in one direction.

The operation of the water tank is modeled in (7)-(12). The state equation of water stored in the tank is formulated in (7), where $\mathbf{V}_t$ is a $(T \times 1)$ dimensional vector of water stored in the tank. The initial water level is set in (8), where $\mathbf{V}_{ini}$ is the initial volume of water in the tank. The pressure head at the tank node is calculated in (9) using the available water in the tank. The available water in the tank is limited by (10) where $\underline{\mathbf{V}}$ is the lower capacity of the tank and $\overline{\mathbf{V}}$ is the upper capacity of tank. The input/output water flow to/from the water tank is limited by (11)-(12), where $\underline{\mathbf{C}}$ and $\overline{\mathbf{C}}$ are lower and upper rates of water flow to tank and $\underline{\mathbf{D}}$ and $\overline{\mathbf{D}}$ are lower and upper rates of water discharge from the tank.

By releasing the stored water in the tank, the hydropower generator of PSH unit can produce electric power according to (13) [25], where η^H is turbine efficiency, $\mathbf{H}_t^{av}$ is $(M \times 1)$ vector of the available head and $\mathbf{Q}_t^{PSH}$ is $(M \times 1)$ vector of the water flow used for power generation. The empirical Hazen-Williams head loss formulation, (14)-(17), is used to model pressure loss in the pipes [26], where $\mathbf{G}$ is an $(S \times S)$ dimensional matrix of the pipe resistance parameters and $\Pi(\cdot)$

is the unit sign function. The available pressure head in each node is limited by (15), where $\underline{\mathbf{H}}$ and $\overline{\mathbf{H}}$ are lower and upper limits of the nodes pressure head. The pressure head is fixed in the reservoir nodes in (16), where $\mathbf{Z}$ is an $(R \times 1)$ dimensional vector of reservoir nodes' elevation. Finally, the volumetric flow rate in WDS is limited by (17), where $\underline{\mathbf{Q}}$ and $\overline{\mathbf{Q}}$ are lower and upper limits of water flow. A linearized form of (3) and (14) is used for reducing the computational burden of this implementation [23].

III. NUMERICAL STUDIES

The proposed model for optimal integration of small PSH units in WDS operation is implemented on a 16-node test WDS, which is shown in Fig. 3. As depicted, this system includes a small PSH unit, water pumping facilities and water tanks. Together, the system components are designed and chosen to help WDS operator optimally deliver water to consumers at the minimum operation cost. The electricity price data are shown in Fig. 4. The 16-node test WDS data are provided in [27].

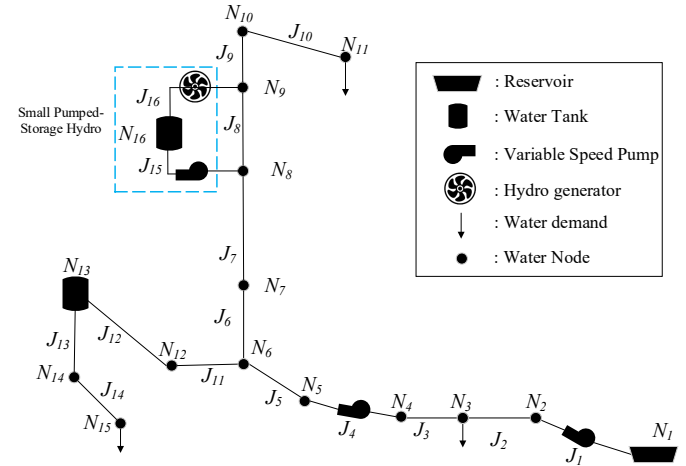


Fig. 3: 16-Node test water distribution system

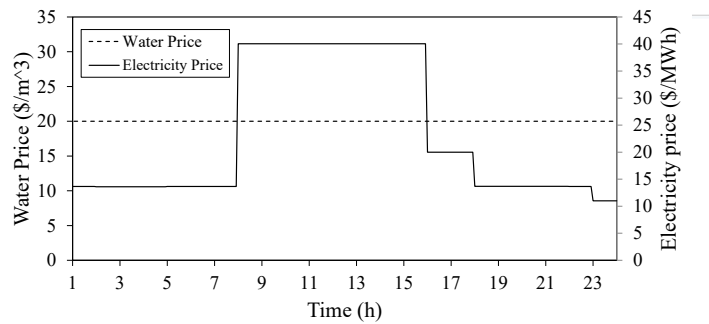


Fig. 4: Input power price

PSH units are composed of different components, including a pumping motor, water tank, and hydropower generator. Based on the expectations from the optimization processes and the application of PSH technology, these units can be configured for operation in different modes. In fact, some

motors used for pumping are reversible and can be converted to generate power as well. Similarly the hydropower generator can be designed for specific operating characteristics. PSH unit, including the hydraulic network, can also be designed to meet specific technical requirements. In this paper, two cases are analyzed to determine the effect of the integration of small PSH units in WDS operation. In Case 1, PSH pumping is used to store water in tanks; however, no power generation is considered from the tank water discharge. On the other side, the effect of power generation obtained from water discharge is included in Case 2. The proposed MINLP model is implemented in GAMS environment, and DICOPT solver is used to solve the model in both cases, on a desktop computer with a 4.0-GHz i7 processor and 32 GB of RAM. The solution times in Cases 1 and 2 are respectively 143.318 and 156.658 seconds.

The total operating cost of WDS in Case 1 is \$152.597, which includes \$81.227 for operation of PSH unit and pumps and \$71.370 for water procurement from water network. In Case 2, total operating cost of the system is \$145.389, which includes \$88.086 for operation of PSH unit and pumps, \$71.370 for water procurement from water reservoirs and \$14.067 for revenue of generating power by PSH unit. These results are associated with the operating modes of available units in WDS that are optimized in Cases 1 and 2 for optimal operation of the system.

Flow rate of water purchased from reservoir node in Cases 1 and 2 are illustrated in Fig. 5 and the power consumption of pumps and small PSH in the mentioned cases is shown in Fig. 6. In both cases, water purchased from the reservoir node increases during the off-peak hours. Specifically for Case 2, the amount of water is slightly greater, compared to Case 1. As electricity prices are lower during these hours, the pumps operate to provide enough pressure head along the pipeline connection, and ensure the storage of water in the available tanks. The water is discharged during the peak hours to supply the water demand with a lower power consumption. The additional amount of purchased water leads to a revenue during the PSH operation in Case 2, which mitigates the pumping overcosts during the charging process and reduces the total operation cost. When enough storage is ensured, power consumption of pumps and small PSH is reduced, as the charged tanks provide water to the demand and pressure head to guarantee WDS operation.

The charge/discharge water flows to/from the small PSH tank are shown in Fig. 7. As mentioned earlier, the water tank is charged during the off-peak period, with an increased flow for Case 2. Once electricity price increases, the operating mode of PSH changes to discharge water to WDS. As additional water was stored in Case 2 in the initial hours, the small PSH provides water for a longer period, compared to Case 1. This storage effect can be related to the revenue received from power generated by the small PSH discharged water, and represents a benefit for WDS operator.

Finally, the consumed/generated power by the small PSH for Cases 1 and 2 is shown in Fig. 8. As the benefit of discharging

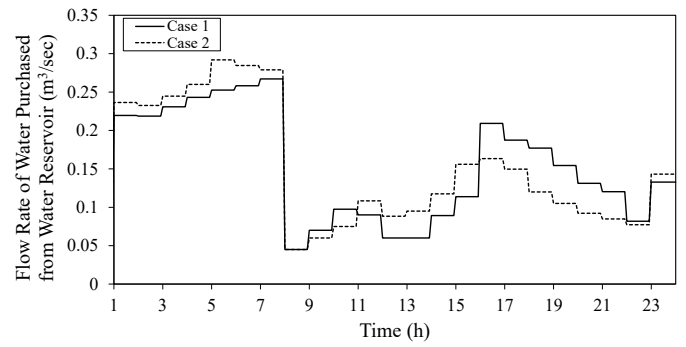


Fig. 5: Flow rate of water purchased from reservoir

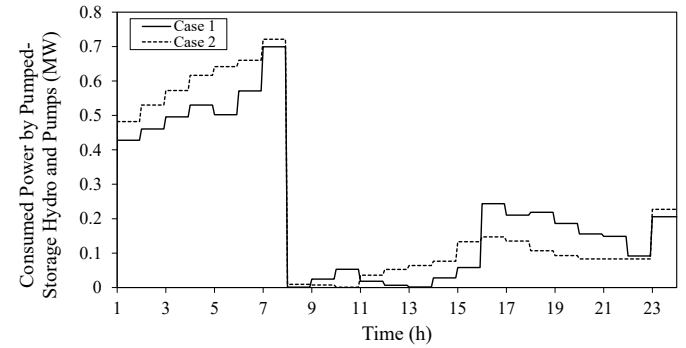


Fig. 6: Power consumption of small PSH and pumps

water is limited to WDS operating condition with no revenue received from power generation, the peak power consumption for pumping purposes in the PSH is reduced by 72% for Case 1, in comparison to Case 2. In Case 2, the water discharge is used for power generation in peak periods, which results in an optimal performance for WDS in terms of economic operation.

IV. CONCLUSION

Energy storage systems have great potential to be integrated in existing systems that utilize water and power. The best-suited storage technology will differ between applications, and depend on the existing infrastructure, operations, and constraints. Small PSH technologies are an appropriate storage mechanism for a diverse set of WDSs, including municipal supplies, irrigation infrastructure, treatment facilities, and buildings. This paper presents an optimization model for integration of small PSH units into WDS's operation. The proposed model satisfies hydraulic constraints of WDS and optimizes energy consumption using small PSH operation modes, i.e., pumping, storage, and generation.

The simulation results on a test 16-node WDS demonstrate that the integrated small PSH unit can successfully improve a WDS operation efficiency through coordinating WDS and PSH operations. The small PSH unit is operated in response to the electricity prices, resulting in optimized power and water consumption costs for WDS. Additionally, other available units in WDS, like variable speed pumps, interact well with

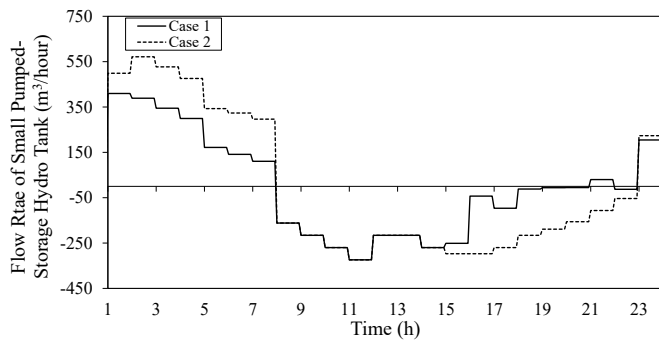


Fig. 7: Flow rate of Small PSH tank

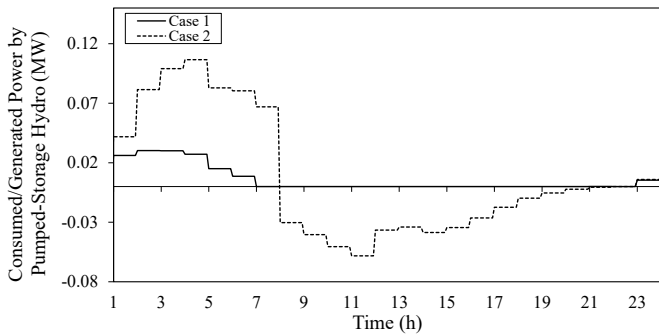


Fig. 8: Consumed/generated power by small PSH unit

the operation of small PSH unit, ensuring reliable and stable operation of the modified WDS.

The proposed optimization approach is directly applicable to a variety of real-world water and energy systems. Future work will demonstrate the role of small PSH units to enhance revenue and flexibility in more complex and specific applications. These techniques can also be analyzed in the context of balancing variable renewable generation profiles, such as wind and solar.

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